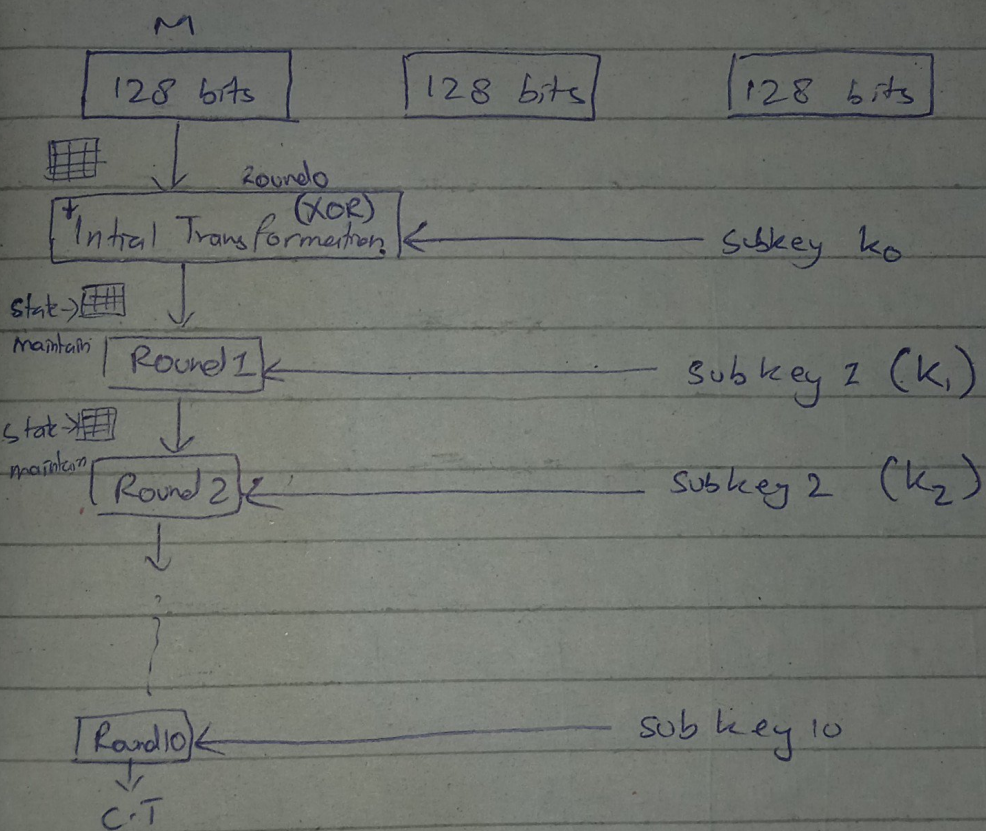


~~ADD~~ Round key

AES (Advance Encryption Standard)

- Block cipher
- Symmetric

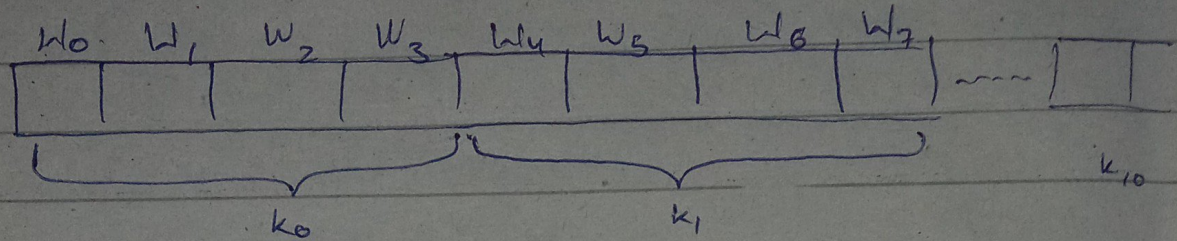


Initial key $\rightarrow$

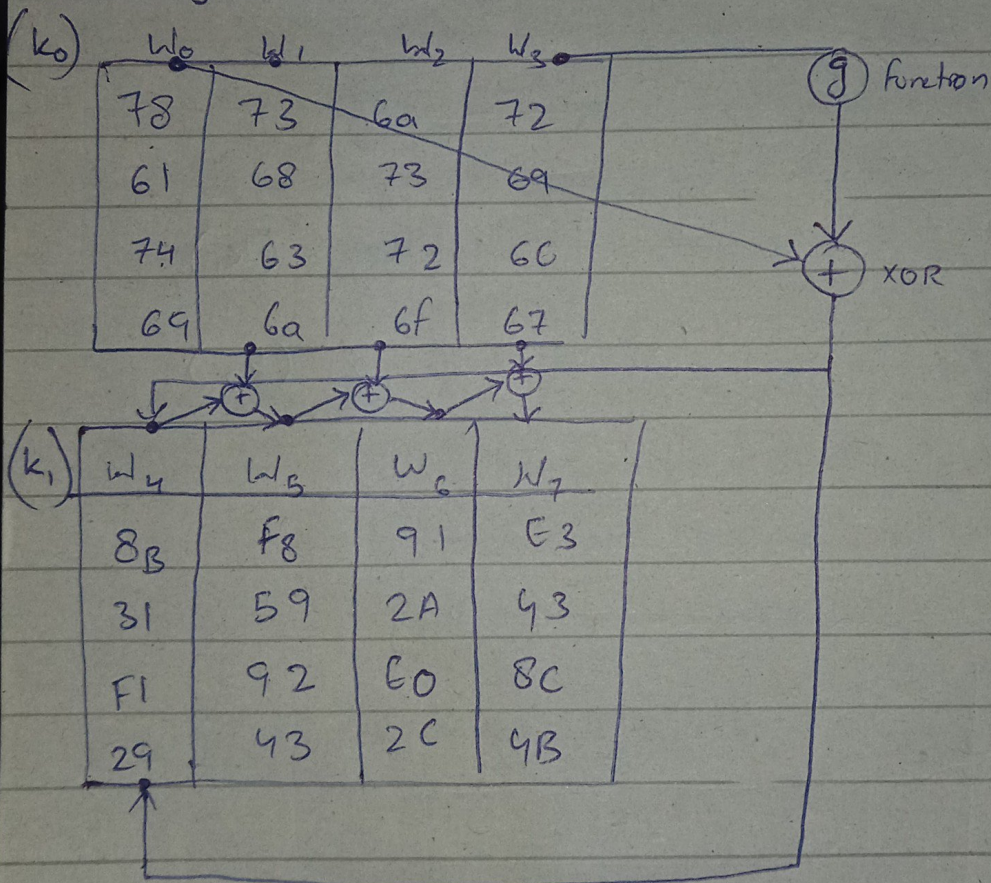
	w_0	w_1	w_2	w_3	
	b_1	b_5	b_9	b_{13}	
	b_2	b_6	b_{10}	b_{14}	
	b_3	b_7	b_{11}	b_{15}	
	b_4	b_8	b_{12}	b_{16}	
	32 bits	32	32	32	= 128 bits

To generate subkeys 1-10 we want key expansion,
 02 - initial key

key Expansion



Initial key



$$\begin{aligned}
 W_4 &= g(W_3) \oplus W_0 \\
 W_5 &= W_4 \oplus W_1 \\
 W_6 &= W_5 \oplus W_2 \\
 W_7 &= W_6 \oplus W_3
 \end{aligned}$$

→ Rotate word

(72) 69 6C 67 ←

→ Sub Word

S-box given

Calculate y (W_3)

W_3	X_1 Rotate word	Y_1 Sub Word	R (const)	R (cant) $\oplus$ Y_1
72	69	F9	01	F8
69	6C	50	00	50
6C	67	85	00	85
67	72	40	00	40

R_1	R_2	R_3	R_4	R_5	R_6	R_7	R_8	R_9	R_{10}
01	02	04	08	10	20	40	80	¹⁸ 16	36
00	00	00	00	00	00	00	00	00	00
00	00	00	00	00	00	00	00	00	00
00	00	00	00	00	00	00	00	00	00

$$\begin{array}{cccccccc}
 & F & 9 & & 5 & 0 & & 8 & 5 \\
 & 1 & 1 & 1 & 1 & 00 & 1 & 01010000 & 10000101 \\
 \oplus & 00 & 00 & 00 & 00 & 00 & 1 & 00000000 & 00000000 \\
 & 1 & 1 & 1 & 1 & 000 & & 01010000 & 10000101
 \end{array}$$

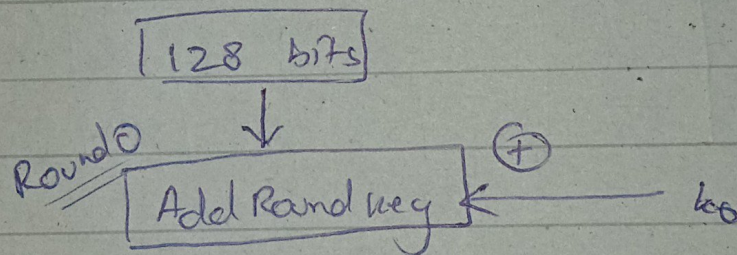
$$\begin{array}{cccc}
 & 4 & & 0 \\
 01 & 00 & 0000 & \\
 00 & 00 & 0000 & \\
 01 & 00 & 0000 &
 \end{array}$$

$R(\text{count}) \oplus Y1 \oplus W0$

$$\begin{array}{r}
 11111000 \quad 01010000 \quad 10000101 \\
 \oplus 01110011 \quad 01100001 \quad 01110100 \\
 \hline
 W_4 \leftarrow 10001011 \quad 00110001 \quad 11110001 \\
 \quad \quad \quad 8 \quad \quad \quad 3 \quad \quad \quad 1 \quad \quad \quad F \quad \quad \quad 1
 \end{array}$$

$$\begin{array}{r}
 01000000 \\
 \oplus 01101001 \\
 \hline
 00101001 \rightarrow W_4 \\
 \quad \quad \quad 2 \quad \quad \quad 9
 \end{array}$$

Example



128 bits

43	11	23	03
23	12	01	03
11	04	61	02
11	01	01	01

2A			

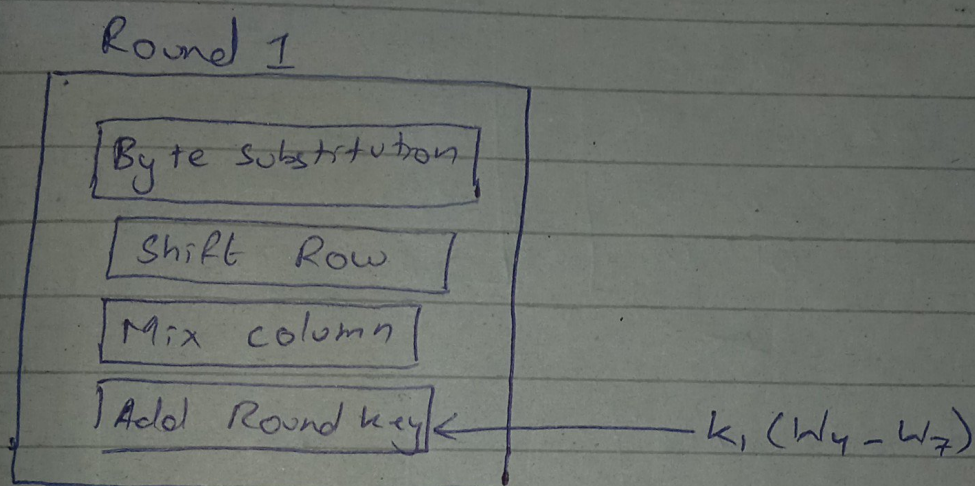
128 bits given

K0

69	11	12	67
43	21	13	70
2F	3F	1F	7B
1B	0F	2F	7A

8421 8421
 0100 0011
 $(7) \quad 0110$ 1010

 0010 1010
 2 A



→ Byte Substitution (step 1)

Output of Round 0

00	16	19	17
04	1C	00	07
17	0e	03	01
16	0F	0F	10

From S-box

63	47	A2	F0
F2	9C	63	C5
F0	A6	7B	7C
9F	76	76	CA

→ Row shift (step 2)

Rule

Row 1 = O-LS

Row 2 $\leftarrow$ 1-LS

Row 3 = 2-LS

Row 4 = 3-LS

8 4 2 1

63	47	A2	F0
9C	63	C5	F2
7B	7C	F0	AB
CA	9F	76	76

Step 3 Mix column

$$\begin{matrix} \text{given} & \xrightarrow{\quad} \\ \begin{bmatrix} 02 & 03 & 01 & 01 \\ 01 & 02 & 03 & 01 \\ 01 & 01 & 02 & 03 \\ 03 & 01 & 01 & 02 \end{bmatrix} & \times & \begin{bmatrix} 63 & 47 & A2 & F0 \\ 96 & 63 & C5 & F2 \\ 7B & 7C & F0 & AB \\ CA & 9F & 76 & 76 \end{bmatrix} \end{matrix}$$

$$\begin{matrix} \text{given} \\ \begin{bmatrix} r_1 & r_5 & r_9 & r_{13} \\ r_2 & r_6 & r_{10} & r_{14} \\ r_3 & r_7 & r_{11} & r_{15} \\ r_4 & r_8 & r_{12} & r_{16} \end{bmatrix} \end{matrix}$$

$$r_1 = (02 \times 63) + (03 \times 9C) + (01 \times 7B) + (01 \times CA)$$

$$r_2 = (01 \times 63) + (02 \times 9C) + (03 \times 7B) + (01 \times CA)$$

$$r_1 = (02 \times 63) + (03 \times 9C) + (01 \times 7B) + (01 \times CA)$$

$$(000000010) (01100011) +$$

$$(00000001) (10011100) +$$

$$(000000001) (0111011) +$$

$$(00000001) (11001010)$$

~~GF~~ GF (2⁸)

Polynomial Arithmetic

$$x^7 + x^6 + x^5 + x^4 + x^3 + x^2 + x + 1$$

$$r_1 = x(x^6 + x^5 + x + 1) + (x+1)(x^7 + x^4 + x^3 + x^2) + (1)(x^6 + x^5 + x^4 + x^3 + x + 1) + (1)(x^7 + x^6 + x^3 + x)$$

A2 F0

CBS F2

FO AB

76 76

$$r_1 = \cancel{x^7} + \cancel{x^6} + \cancel{x^2} + \cancel{x} + \cancel{x^8} + \cancel{x^4} + \cancel{x^3} + \cancel{x^7} + \cancel{x^4} + \cancel{x^3} + \cancel{x^2} + \cancel{x^6} + \cancel{x^5} + \cancel{x^4} + \cancel{x^3} + \cancel{x} + 1 + \cancel{x^7} + \cancel{x^6} + \cancel{x^3} + \cancel{x}$$

= 2

$$r_1 = x^8 + x^7 + x^6 + x^4 + x + 1$$

→ two same

power cancel out

1 1 1 0 1 0 0 1 1

9 bits out of 2⁸

3) + (01 x CA

B) + (01 x

(01 x CA)

if 9 bits we will do Reducible Polynomial Equation

$$P(x) = x^8 + x^4 + x^3 + x + 1$$

$$P(x) = 100011011$$

→ remember

P(x)

$$100011011 \begin{array}{r} 111010011 \\ 100011011 \\ \hline 011001000 \end{array}$$

cancel out

←

$$011001000$$

A-10
B-11
C-12

8421

11 00 1000
B 8

$$r_2 = (01 \times 63) + (02 \times 9C) + (03 \times 7B) + (01 \times CA)$$

$$\begin{aligned} & (0000 \ 0001) (0110 \ 0011) + \\ & (0000 \ 0010) (01001 \ 1100) + \\ & (0000 \ 0011) (0111 \ 1010) + \\ & (0000 \ 0001) (1100 \ 1010) \end{aligned}$$

$$x^7 + x^6 + x^5 + x^4 + x^3 + x^2 + x + 1$$

$$\begin{aligned} & (1) (x^6 + x^5 + x + 1) + (x) (x^7 + x^4 + x^3 + x^2) \\ & + (x+1) (x^6 + x^5 + x^4 + x^3 + x + 1) + \\ & (1) (x^7 + x^6 + x^3 + x) \end{aligned}$$

$$\rightarrow \gcd(19, 72)$$

$$72 = 19 \times 3 + 15$$

$$19 = 15 \times 1 + 4$$

$$15 = 4 \times 3 + 3$$

$$4 = 2 \times 2 + 0$$

$\rightarrow$ Congruence

$$a \equiv b \pmod{m}$$

$$46 \equiv 68 \pmod{11}$$

$$a \pmod{m} = b \pmod{m}$$

$$46 \pmod{11} = 68 \pmod{11}$$

$\rightarrow$ Chinese Remainder Theorem

$$x \equiv 2 \pmod{3}$$

$$x \equiv a_1 \pmod{m_1}$$

$$x \equiv 3 \pmod{5}$$

$$x \equiv a_2 \pmod{m_2}$$

$$x \equiv 2 \pmod{7}$$

$$x \equiv a_3 \pmod{m_3}$$

$$x \equiv a_n \pmod{m_n}$$

$$x \equiv (M_1 x_1 a_1 + M_2 x_2 a_2 + M_3 x_3 a_3 + \dots + M_n x_n a_n) \pmod{M}$$

$$\pmod{M}$$

$$M = m_1 \times m_2 \times m_3$$

$$M_1 = \frac{M}{m_1} = \frac{m_1 \times m_2 \times m_3}{m_1} = m_2 \times m_3$$

$$M_2 = m_1 \times m_3$$

$$M_3 = m_1 \times m_2$$

$$M_1 x_1 \equiv 1 \pmod{m_1}$$

$$M_2 x_2 \equiv 1 \pmod{m_2}$$

$$M_3 x_3 \equiv 1 \pmod{m_3}$$

if not 1 we stop and don't solve further
because numbers are not relatively prime
Hence, congruence cannot be ~~find~~

$$\gcd(m_1, m_2) = \gcd(m_2, m_3) = \gcd(m_3, m_1) = 1$$

$$\begin{aligned} Q) \quad x &\equiv 1 \pmod{5} \\ x &\equiv 1 \pmod{7} \\ x &\equiv 3 \pmod{11} \end{aligned}$$

$$\gcd(m_1, m_2)$$

$$7 = 5 \times 1 + 2$$

$$5 = 2 \times 2 + 1 \leftarrow$$

$$2 = 1 \times 2 + 0 \leftarrow$$

$$\gcd(7, 11)$$

$$11 = 7 \times 1 + 4$$

$$7 = 4 \times 1 + 3$$

$$4 = 3 \times 1 + 1 \leftarrow$$

$$3 = 1 \times 3 + 0 \leftarrow$$

$$\gcd(5, 11)$$

$$11 = 5 \times 2 + 1 \leftarrow$$

$$5 = 1 \times 5 + 0 \leftarrow$$

~~2~~

$$M = m_1 \times m_2 \times m_3 = 5 \times 7 \times 11 = 385$$

$$M_1 = m_2 m_3 = 77$$

$$M_2 = m_1 m_3 = 55$$

$$M_3 = m_1 m_2 = 35$$

$$\begin{array}{r} 15 \\ 5 \overline{) 77} \\ \underline{- 75} \\ 2 \end{array}$$

$$M_1 x_1 \equiv 1 \pmod{m_1}$$

$$77 x_1 \pmod{5} = 1 \pmod{5}$$

$$2 x_1 \pmod{5} = 1$$

$$x_1 = 3$$

↳ guess it such that remainder = 1

$$\begin{array}{r} 7 \\ 7 \overline{) 55} \\ \underline{- 49} \\ 6 \end{array}$$

$$M_2 x_2 \equiv 1 \pmod{7}$$

$$55 x_2 \pmod{7} = 1$$

$$6 x_2 \pmod{7} = 1$$

$$x_2 = 6$$

$$\begin{array}{r} 5 \\ 7 \overline{) 36} \\ \underline{- 35} \\ 1 \end{array}$$

Testing

further
prime

$$M_3 x_3 \equiv 1 \pmod{m_3}$$

$$35 x_3 \pmod{11} \equiv 1 \pmod{11}$$

$$2 x_3 \pmod{11} \equiv 1$$

$$x_3 \equiv 6$$

$$\begin{array}{r} 1 \\ 11 \overline{) 12} \\ \underline{- 11} \\ 1 \end{array}$$

$$x \equiv (M_1 x_1 a_1 + M_2 x_2 a_2 + M_3 x_3 a_3 + \dots + M_n x_n a_n) \pmod{M}$$

5, 11)

2 + 1

5 + 10

$$x \equiv (77 \times 3 \times 1) + (55 \times 6 \times 1) + (35 \times 6 \times 3) \pmod{385}$$

$$x \equiv 1191 \pmod{385}$$

$$\boxed{x = 36}$$

Testing

$$36 \pmod{5} = 1 \pmod{5}$$

$$36 \pmod{7} = 1$$

$$36 \pmod{11} = 3$$

$$\begin{array}{r} 3 \\ 385 \overline{) 1191} \\ \underline{- 1155} \\ 36 \end{array}$$

Information Security

→ Linear Congruential Generator

$$X_{n+1} = (a X_n + c) \bmod m$$

$$X_1 = (a X_0 + c) \bmod m$$

$$X_2 = (a X_1 + c) \bmod m$$

$$X_3$$

}

$$X_{n+1}$$

Q) Generate random number and tell its period

→ Blum Blum Shub Generator

$$p \equiv q \equiv 3 \pmod{4}$$

$$p \bmod 4 = 3$$

Stream cipher

Stream cipher

Key Initial = [1 2 3 6]

$S = [0 \ 1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7]$
 $T = [1 \ 2 \ 3 \ 6 \ 4 \ 5 \ 7]$

• Now perform initial permutation on S

$j = 0$

for $i = 0$ to 7

do $j = (j + S[i] + T[i]) \bmod 8$

swap ($S[i]$, $S[j]$);

end

$\rightarrow i = 0, j = 0$

$j = (j + S[0] + T[0]) \bmod 8$

$j = (0 + 0 + 1) \bmod 8$

$j = 1$

Swap $S[0]$, $S[1]$

$S = [1 \ 0 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7]$

$\rightarrow i = 1, j = 1$

$j = (1 + 0 + 2) \bmod 8$

$j = 3$

swap $S[1], j[3]$

$S = [1 \ 3 \ 2 \ 0 \ 4 \ 5 \ 6 \ 7]$

{
,
,
,
,
,
}

$S = \begin{matrix} & 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ [2 & 3 & 7 & 4 & 6 & 0 & 1 & 5] \end{matrix}$

$T = [1 \ 2 \ 3 \ 6 \ 1 \ 2 \ 3 \ 6]$

Now

$i, j = 0;$

while (true)

{

$i = (i+1) \bmod 8;$

$j = (j + S[i]) \bmod 8;$

swap ($S[i], S[j]$);

$t = (S[i] + S[j]) \bmod 8;$

$K = S[t];$

}

$t = 4 + 3$

$t = 7$

plantent $[\textcircled{1} 2 2 2]$

$$K = [5 \quad _ \quad _ \quad _]$$

$$001 \rightarrow 1$$

$$\textcircled{+} 101 \rightarrow 5$$

$$100$$

$$\hookrightarrow 4$$

$$C.T = 4$$

$$C.T = [4 \quad _ \quad _ \quad _]$$

Diffie-Hellman Key Agreement

$\{1, 2, 3, 4, 5, 6\}$

③ $q = 7$
 $\alpha = ?$

should be the
Primitive root of q

$n = \alpha^m \bmod q$ let $\alpha = 2$

$2^1 \bmod 7 = 2$

$2^2 \bmod 7 = 4$

$2^3 \bmod 7 = 1$

$2^4 \bmod 7 = 2$

$\times$ 2^2 cannot come
again so 2 is not
primitive root

let $\alpha = 3$

$3^1 \bmod 7 = 3$

$3^2 \bmod 7 = 2$

$3^3 \bmod 7 = 6$

$3^4 \bmod 7 = 4$

$3^5 \bmod 7 = 5$

$3^6 \bmod 7 = 1$

3 is primitive root, another number can
also be primitive root other than 3

④ $q = 11$
 $\alpha = ?$

Authentication

MAC diagrams



All